ex. Find two linearly independent solutions to the equation:

1. $t^2y'' + 3ty' - 3y = 0$ $a=1$ $b=3$ $c=-3$

Char. eqn: $r^2 + (3-1)r - 3 = 0$

$r^2 + 2r - 3 = 0$

$(r+3)(r-1) = 0$ $r = 1, -3$

$y_1 = t$ $y_2 = t^{-3}$

2. $t^2y'' + 3ty' + y = 0$ $a=1,$ $b=3,$ $c=1$

$r^2 + (3-1)r + 1 = 0$

$r^2 + 2r + 1 = 0$

$(r+1)^2 = 0$ $r=-1$ is a double root

$y_1 = t^{-1}$ $y_2 = t^{-1}\ln t$

3. $3t^2y'' + ty' + 2y = 0$ $\quad a=3 \quad b=1 \quad c=2$

$3r^2 + (1-3)r + 2 = 0$

$3r^2 - 2r + 2 = 0$

$$r = \frac{2 \pm \sqrt{4-24}}{6} = \frac{1}{3} \pm \frac{i\sqrt{5}}{3}$$

$$y_1 = t^{1/3} \cos\left(\frac{\sqrt{5}}{3} \ln t\right)$$

$$y_2 = t^{1/3} \sin\left(\frac{\sqrt{5}}{3} \ln t\right)$$